Graphing Rational Functions

Section 1: Multiplicity of Zeros and Sign Analysis

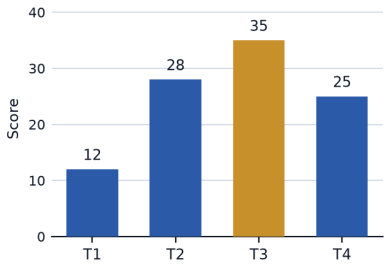
1 If a factor $(x - c)$ in the denominator of a rational function has odd multiplicity (e.g. $(x - c)^1$), how does the graph behave near the asymptote $x = c$?

a. The graph goes in opposite directions ($+\infty$ on one side and $-\infty$ on the other)

b. The graph goes in the same direction ($+\infty$ on both sides)

c. The graph crosses the axis

d. A hole is formed



2 If the factor in the denominator has even multiplicity $(x - c)^2$, how does the graph behave near $x = c$?

a. The graph goes in the same direction on both sides (both $+\infty$ or both $-\infty$)

b. The graph goes in opposite directions

c. The graph crosses the asymptote

d. A corner is formed

3 What is the sign behavior of $f(x) = \frac{1}{(x-2)^2}$ on both sides of $x = 2$?

a. Positive ($+\infty$) on both sides

b. Negative on left, positive on right

c. Negative on both sides

d. Undefined

Section 2: Constructing Sign Charts

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4 Which values partition the number line into test intervals when making a sign chart for a rational function?

a. All real zeros of the numerator and all real zeros of the denominator

b. Only the zeros of the denominator

c. Only the zeros of the numerator

d. Only the y-intercept

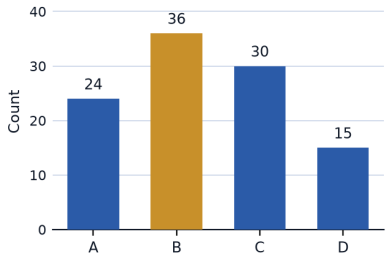
5 Find the solution set of $\frac{x-3}{x+1} > 0$.

a. $(-\infty, -1) \cup (3, \infty)$

b. $(-1, 3)$

c. $[3, \infty)$

d. $(-\infty, 3)$



6 Solve the inequality $\frac{x^2-4}{x} \leq 0$.

a. $(-\infty, -2] \cup (0, 2]$

b. $[-2, 0) \cup [2, \infty)$

c. $[-2, 2]$

d. $(-\infty, 0)$

Section 3: Sketching Complete Curves

7 Which of the following is the complete set of features needed to accurately sketch a rational function?

a. Domain, holes, vertical/horizontal asymptotes, x- and y-intercepts, and sign behavior

b. Only the degree and the leading coefficient

c. Only the y-intercept and one point

d. Only the vertical asymptotes

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8 Sketch $f(x) = \frac{2x}{x-1}$. What is the value of $f(2)$ and $f(-1)$?

a. $f(2) = 4$ and $f(-1) = 1$

b. $f(2) = 2$ and $f(-1) = -1$

c. $f(2) = 4$ and $f(-1) = -1$

d. $f(2) = 1$ and $f(-1) = 2$

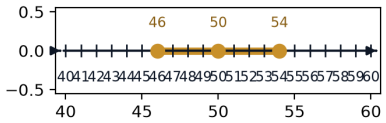
9 In $f(x) = \frac{2x}{x-1}$, as $x \to 1^+$, what is the limit of $f(x)$?

a. $+\infty$

b. $-\infty$

c. 2

d. 0



Section 4: Writing Rational Functions from Features

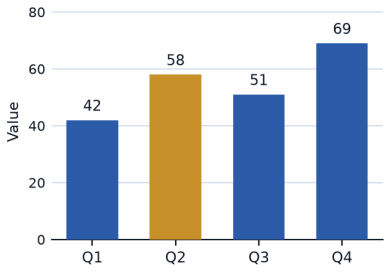
10 Write a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept $(1, 0)$.

a. $f(x) = \frac{x-1}{(x+2)(x-3)}$

b. $f(x) = \frac{(x+2)(x-3)}{x-1}$

c. $f(x) = \frac{x-1}{x^2+1}$

d. $f(x) = \frac{2x-1}{(x+2)(x-3)}$



11 Write a rational function with vertical asymptote $x = 4$, hole at $x = -1$, horizontal asymptote $y = 2$, and y-intercept $(0, -1/2)$.

a. $f(x) = \frac{2(x+1)(x-1)}{(x+1)(x-4)}$

b. $f(x) = \frac{2x+1}{x-4}$

c. $f(x) = \frac{(x+1)}{(x+1)(x-4)}$

d. $f(x) = \frac{2(x-4)}{x+1}$

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12 What is the minimum degree of the denominator of a rational function with two distinct vertical asymptotes?

a. 2

b. 1

c. 3

d. 4

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